

ATHARVA TAYADE

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SUMMARY

Electrical engineer with hands-on power electronics and circuit design work. Designed a solar charge controller end to end, from schematic capture and PCB layout through ESP32 firmware. Experienced in power electronics, including boost converters and MOSFET, with knowledge of PWM, ADC, I2C, and SPI. Familiar with SCADA analysis and smart grid mapping from work on live substations.

EDUCATION

Fr. C. Rodrigues Institute of Technology (FCRIT), Navi Mumbai - University of Mumbai

Bachelor of Engineering (B.E.), Electrical Engineering - May 2026

TECHNICAL SKILLS

Power electronics: Boost converters (CCM/DCM), MPPT (P&O, incremental conductance), MOSFET switching, Gate drive circuits, Resonant WPT

Circuit design: KiCad, Schematic capture, PCB layout, Analog and digital design

Embedded: ESP32, Microcontroller programming, PWM, ADC, I2C, SPI, HCPL-3120, ACS712, IRF540N

Control and simulation: MATLAB/Simulink, PI/PID, Fuzzy Logic, Transient Analysis, Power Factor Correction, Feedback Control

Languages: MATLAB, Python, JavaScript, TypeScript, SQL, Bash, C, C++

Skills: Power Electronics, Wireless Power Transfer

EXPERIENCE

Engineering Intern - Maharashtra State Electricity Board (MSEB)

Navi Mumbai | Dec 2024 - Jan 2025

- Observed day-to-day operations at 33/11 kV distribution substations, from voltage step-down to the practical difficulties of keeping supply stable
- Learned how faults are located and cleared, and how engineers correct power-factor and frequency deviations on the grid
- Studied how the substation is laid out through its single-line diagrams (SLD), along with the on-site operational do's and don'ts and safety practices
- Shadowed technicians through routine transformer and switchgear maintenance

Software Engineering Intern - Sumati.io

Remote | Jan 2024 - Mar 2024

- Rebuilt the policy module UI in React with component-level code splitting; page load time dropped 45%, measured in Lighthouse
- Added response caching to .NET Core API endpoints, cutting redundant backend calls about 60% on highest-traffic policy routes
- Built a monitoring dashboard tracking request latency, error rates, and service health across 4 microservices

PROJECTS

MPPT Solar Charge Controller - Final Year Capstone

ESP32, Boost Converter, P&O, HCPL-3120, KiCad

- Designed a 40 W controller (32.8 V to 48 V, 20 kHz) end to end: schematic capture, PCB layout, and firmware
- Wrote ESP32 firmware running P&O MPPT tracking and PI voltage regulation together
- Measured 90.2% system efficiency and 40.1% power gain over fixed duty cycle

ML-Based MPPT Algorithm Selector

Python, PyTorch, scikit-learn, Fuzzy Logic

- Built a classifier that picks between P&O, incremental conductance, and fuzzy logic MPPT from irradiance and temperature inputs
- Trained on 10,000+ simulated solar profiles; 92% accuracy on a held-out test set with 5-fold cross-validation

Resonant Wireless Power Transfer System - Third Year Project

ESP32, Series-Series Resonant Coupling, 85-150 kHz

- Built a 10-30 W transfer system across a 3-8 cm air gap at 60-80% efficiency
- Hand-wound planar spiral coils (20-100 uH); ESP32 PWM with P&O frequency tuning
- Series-series compensation with MOSFET high-frequency inverter; rectifier and DC-DC for 5 V and 12 V output

HACKATHONS

Smart India Hackathon 2023 - Participant - FCRIT internal round (AICTE, Ministry of Education)

Agnethon 2024 - Participant - CSI Student Chapter, FCRIT